

Exploratory Data Analysis on the Automobile Dataset

Report

Introduction

This report presents an exploratory data analysis (EDA) of the automobile dataset. The dataset contains various attributes of vehicles including price, engine size, fuel type, horsepower, and fuel efficiency. The objective of this analysis is to clean the dataset, explore relationships between variables, and extract meaningful insights.

Data cleaning

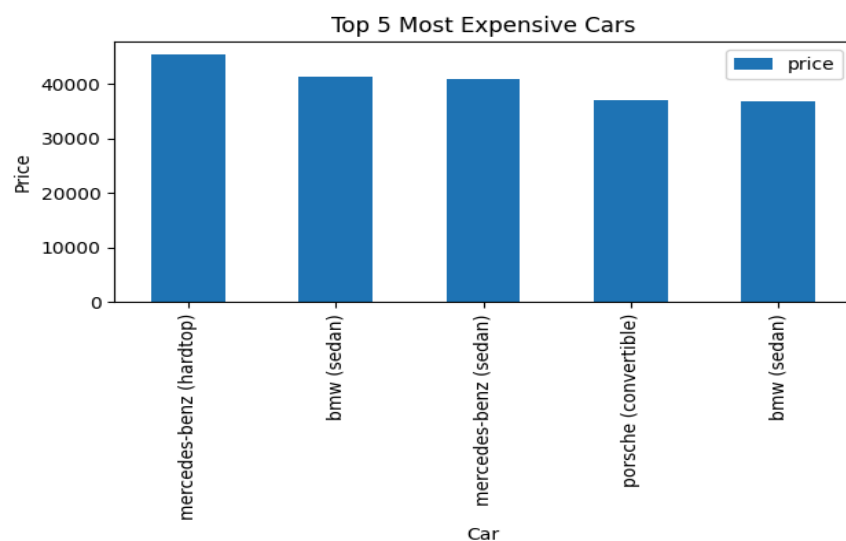
The dataset required several preprocessing steps before analysis. The raw data contained no single value fields, so no fields could be removed based on redundancy. No fully duplicate records were identified either, but to be thorough the steps were implemented to remove any such records. The final step to cleaning the data was converting several, numerical fields to their appropriate data types (int64 and float64).

Missing data

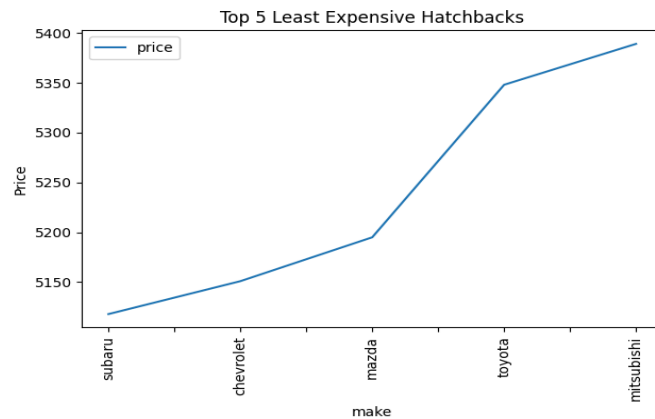
At first glance, there seemed to be no missing data. This was because the raw dataset used "?" to represent missing values instead of standard NaN values. After replacing the question marks with NaN values, there were 59 missing entries in total. By removing the first two fields,, "symbolising" and "normalised-losses", as instructed, 41 missing entries were removed. The remaining 18 missing entries were removed by dropping the records containing the missing data.

Data Stories and visualisations

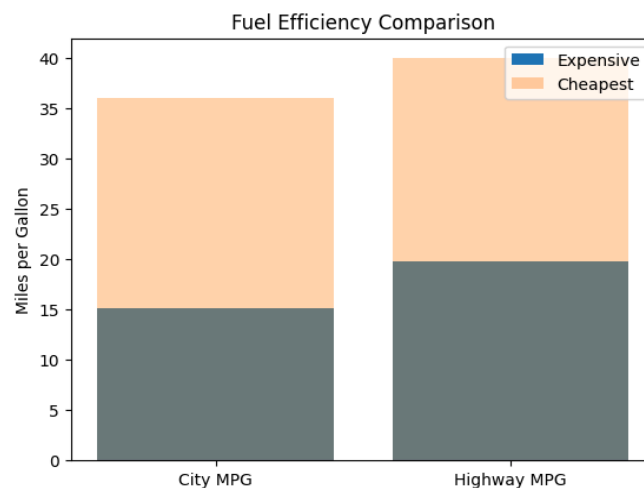
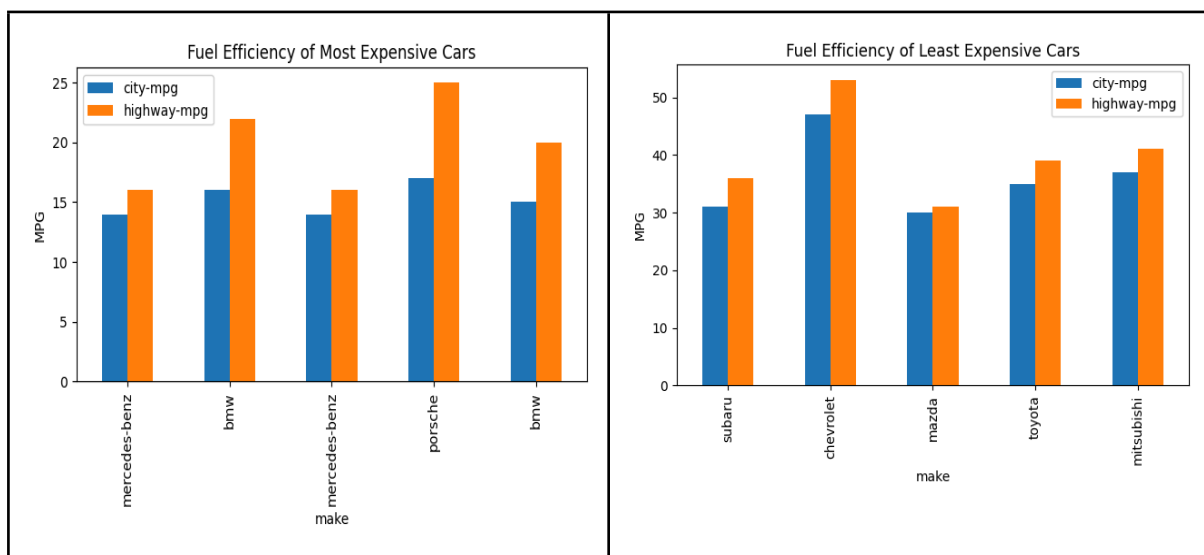
The most expensive vehicles: The five most expensive cars are all made by German automotive manufacturers! As shown below, a Mercedes-Benz hardtop ranks as the expensive vehicle in the dataset.



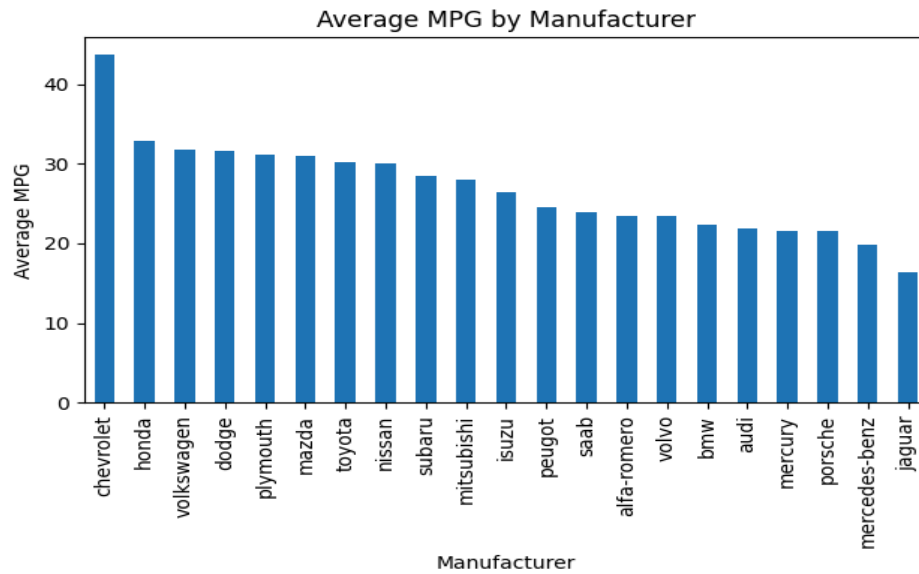
The least expensive vehicles: The five least expensive cars are shown below, and all of them happen to be hatchbacks! A Subaru hatchback is therefore the least expensive of all the cars in the dataset.



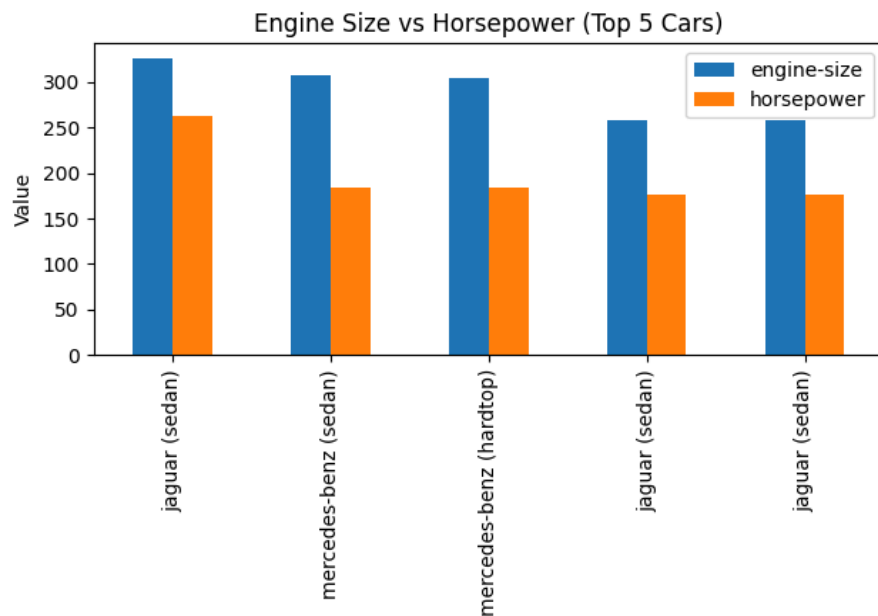
Relationship between price and fuel efficiency (MPG): The fuel efficiency of the five most expensive and the five least expensive cars are shown below, and the results may shock you! The five least expensive cars are nearly twice as cost-effective or fuel-efficient when compared to the five most expensive cars!



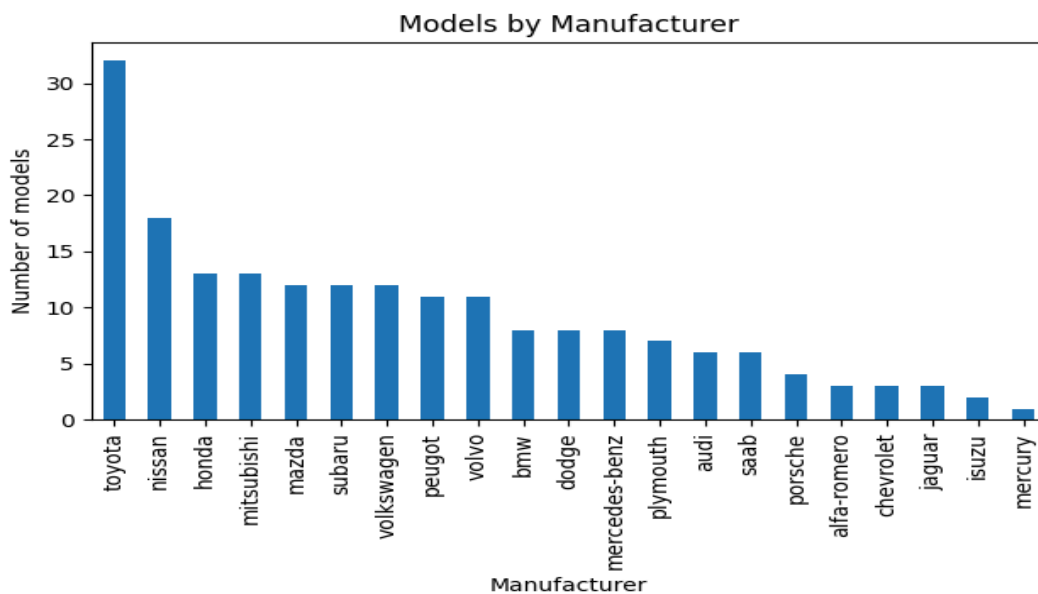
Most fuel-efficient manufacturer: Chevrolet leads the way as the most fuel-efficient automobile manufacturer, while Jaguar tails as the least fuel-efficient manufacturer.



Models with largest engine sizes: Exclusively dominating the top five list for raw power, Jaguar and Mercedes-Benz install the largest engine sizes of all the manufacturers. Jaguar installs the largest of the engines, which also churns the most horsepower of the top five largest engines in the dataset. This observation echoes why Jaguar ranked last in the fuel-efficiency comparison.



Manufacturer with the most models in the dataset: Toyota manufactures the most models or variants of all the manufacturers in the dataset, nearly double the number of models manufactured by Nissan in second place!



Conclusion

The analysis demonstrates a negative relationship between price and fuel efficiency (MPG). Vehicles with larger engine sizes are associated with higher prices and performance. Additional vehicle characteristics such as manufacturer and body type also influence both the vehicle price and the fuel efficiency.

While the most expensive cars offer neater aesthetics, greater power and better performance, they are by no means economical in terms of fuel consumption. Less expensive vehicles generally exhibit higher fuel efficiency, making them the more cost-effective option in terms of fuel consumption.

This analysis provides valuable insights into consumer trade-offs between performance, cost, and economy when selecting vehicles.

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